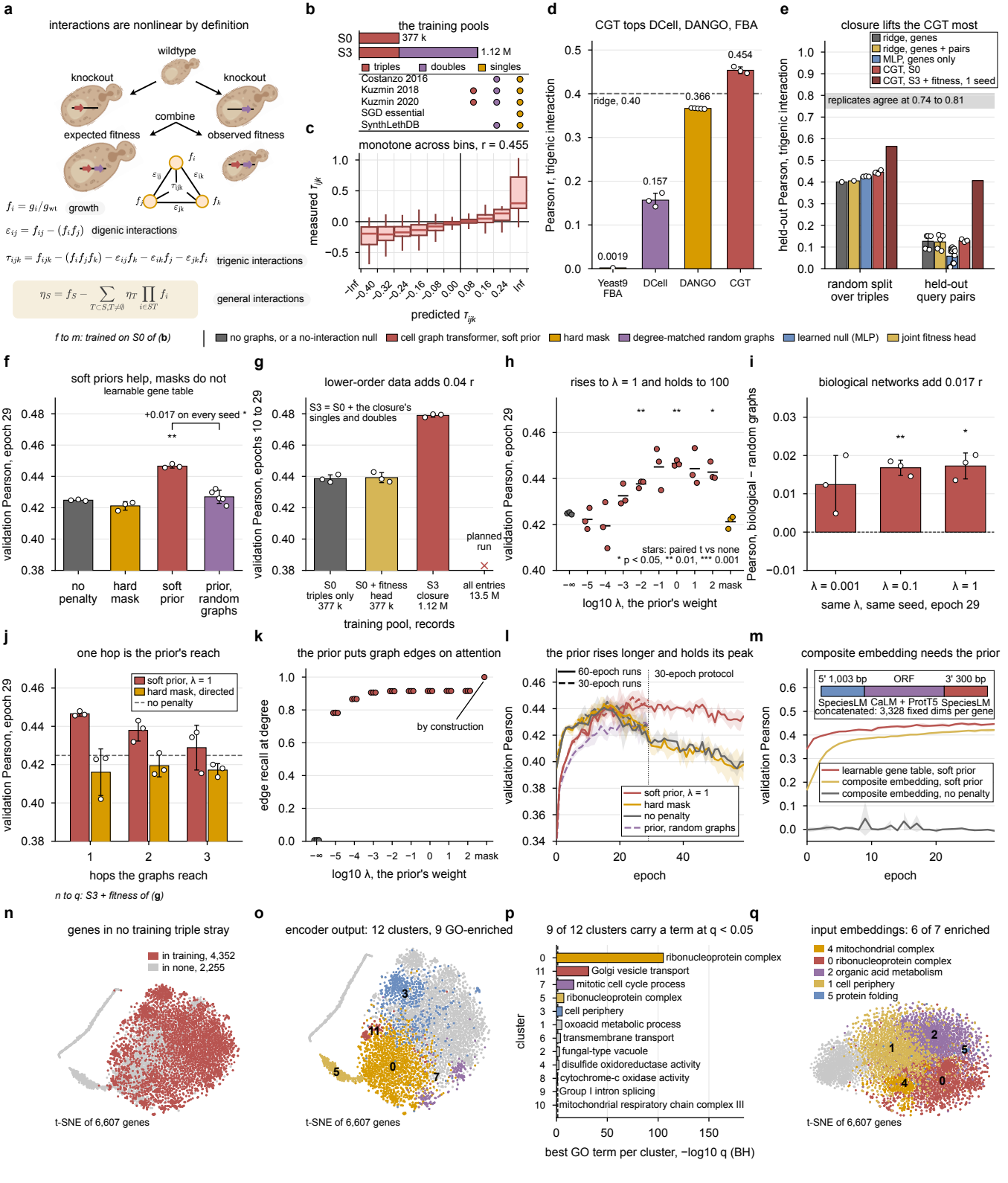

a
interactions are nonlinear by definition
wildtype
knockout
knockout
combine
expected fitness
observed fitness
$f_i = g_i/g_{wt}$ growth
$\varepsilon_{ij} = f_{ij} - (f_i f_j)$ digenic interactions
$\tau_{ijk} = f_{ijk} - (f_i f_j f_k) - \varepsilon_{ij} f_k - \varepsilon_{ik} f_j - \varepsilon_{jk} f_i$ trigenic interactions
$\eta_S = f_S - \sum_{T \subset S, T \neq \emptyset} \eta_T \prod_{i \in ST} f_i$ general interactions
b
the training pools
S0
377 k
S3
1.12 M
triples
doubles
singles
Costanzo 2016
Kuzmin 2018
Kuzmin 2020
SGD essential
SynthLethDB
c
monotone across bins, r = 0.455
measured τ_{ijk}
predicted τ_{ijk}
d
CGT tops DCell, DANGO, FBA
Pearson r, trigenic interaction
ridge, 0.40
0.0019
0.157
0.366
0.454
Yeast9 FBA
DCell
DANGO
CGT
e
closure lifts the CGT most
ridge, genes
ridge, genes + pairs
MLP, genes only
CGT, S0
CGT, S3 + fitness, 1 seed
replicates agree at 0.74 to 0.81
held-out Pearson, trigenic interaction
random split over triples
held-out query pairs
f to m: trained on S0 of (b)
no graphs, or a no-interaction null
cell graph transformer, soft prior
hard mask
degree-matched random graphs
learned null (MLP)
joint fitness head
f
soft priors help, masks do not
learnable gene table
+0.017 on every seed *
**
validation Pearson, epoch 29
no penalty
hard mask
soft prior
prior, random graphs
g
lower-order data adds 0.04 r
S3 = S0 + the closure's singles and doubles
validation Pearson, epochs 10 to 29
planned run
S0 triples only 377 k
S0 + fitness head 377 k
S3 closure 1.12 M
all entries 13.5 M
training pool, records
h
rises to λ = 1 and holds to 100
validation Pearson, epoch 29
stars: paired t vs none
* p < 0.05, ** 0.01, *** 0.001
log10 λ, the prior's weight
i
biological networks add 0.017 r
Pearson, biological − random graphs
λ = 0.001
λ = 0.1
λ = 1
same λ, same seed, epoch 29
j
one hop is the prior's reach
soft prior, λ = 1
hard mask, directed
no penalty
validation Pearson, epoch 29
hops the graphs reach
k
the prior puts graph edges on attention
edge recall at degree
by construction
log10 λ, the prior's weight
l
the prior rises longer and holds its peak
60-epoch runs
30-epoch runs
30-epoch protocol
soft prior, λ = 1
hard mask
no penalty
prior, random graphs
validation Pearson
epoch
m
composite embedding needs the prior
5' 1,003 bp
ORF
3' 300 bp
SpeciesLM
CaLM + ProtT5
SpeciesLM
concatenated: 3,328 fixed dims per gene
learnable gene table, soft prior
composite embedding, soft prior
composite embedding, no penalty
validation Pearson
epoch
n to q: S3 + fitness of (g)
n
genes in no training triple stray
in training, 4,352
in none, 2,255
t-SNE of 6,607 genes
o
encoder output: 12 clusters, 9 GO-enriched
t-SNE of 6,607 genes
p
9 of 12 clusters carry a term at q < 0.05
cluster
ribonucleoprotein complex
Golgi vesicle transport
mitotic cell cycle process
ribonucleoprotein complex
cell periphery
oxoacid metabolic process
transmembrane transport
fungal-type vacuole
disulfide oxidoreductase activity
cytochrome-c oxidase activity
Group I intron splicing
mitochondrial respiratory chain complex III
best GO term per cluster, −log10 q (BH)
q
input embeddings: 6 of 7 enriched
4 mitochondrial complex
0 ribonucleoprotein complex
2 organic acid metabolism
1 cell periphery
5 protein folding
t-SNE of 6,607 genes